

THE FIRST PUBLIC EYE-TRACKING DATASET

of natural-image free viewing through systematically varied polygonal apertures
A controlled testbed for how viewing-region geometry shapes gaze.

30

Participants

600

Images
from CAT2000

6

Different Categories
from CAT2000

27

Polygons

410K

Fixations

Why Polygonal Apertures?

In everyday vision, scenes are **rarely seen whole**.
They appear through windows, doorways, head-mounted displays, and irregular openings.
All existing saliency datasets use **full, rectangular stimuli**.
No Public Dataset systematically varies aperture geometry.
PolyCAT is a **controlled testbed** for how viewing constraints reshape visual exploration.

Dataset at a Glance

Apparatus EyeLink 1000+ (head-mounted)
Display 27" 4K @ 144 Hz
Viewing 70 cm · ~78.5 px/° VA
Sampling 500 Hz binocular
Calibration 13-point + per-trial drift correction
Trials 21,026 total — 97.2% valid
Fixations 19.6 / trial · mean 322 ms
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Data Pipeline

EDF recording

EyeLink raw

Sample streams

TSV · 83.8 M samples

Fixation events

CSV · 410,166

Saliency maps

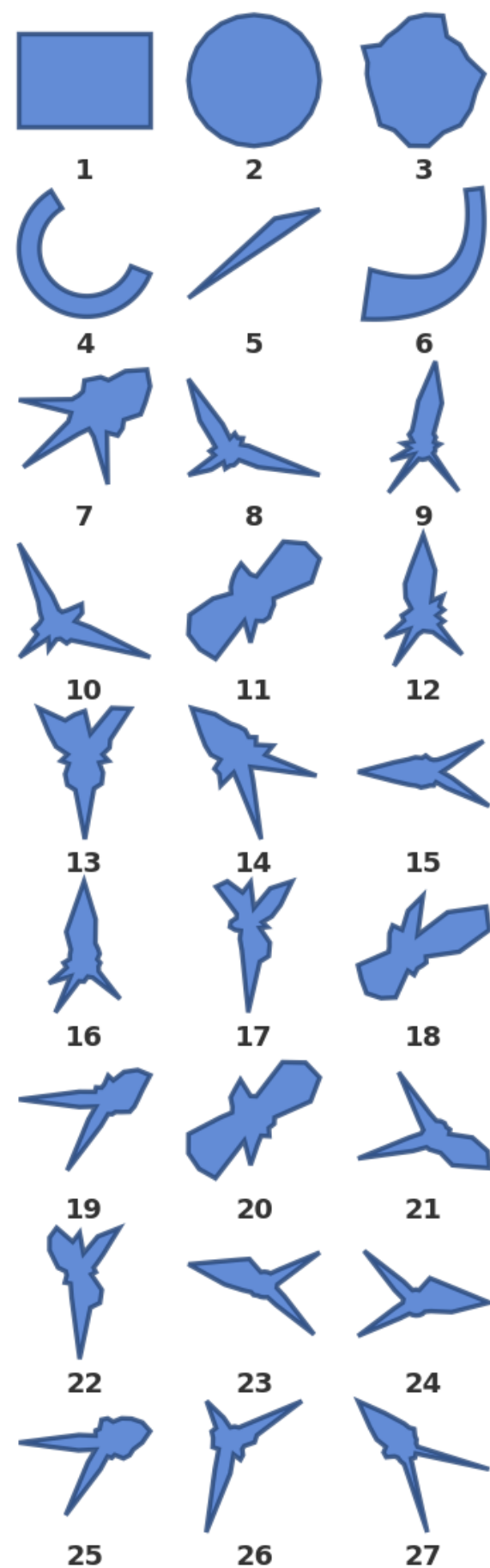
NumPy / PNG · 648

Benchmarks

MIT metrics suite

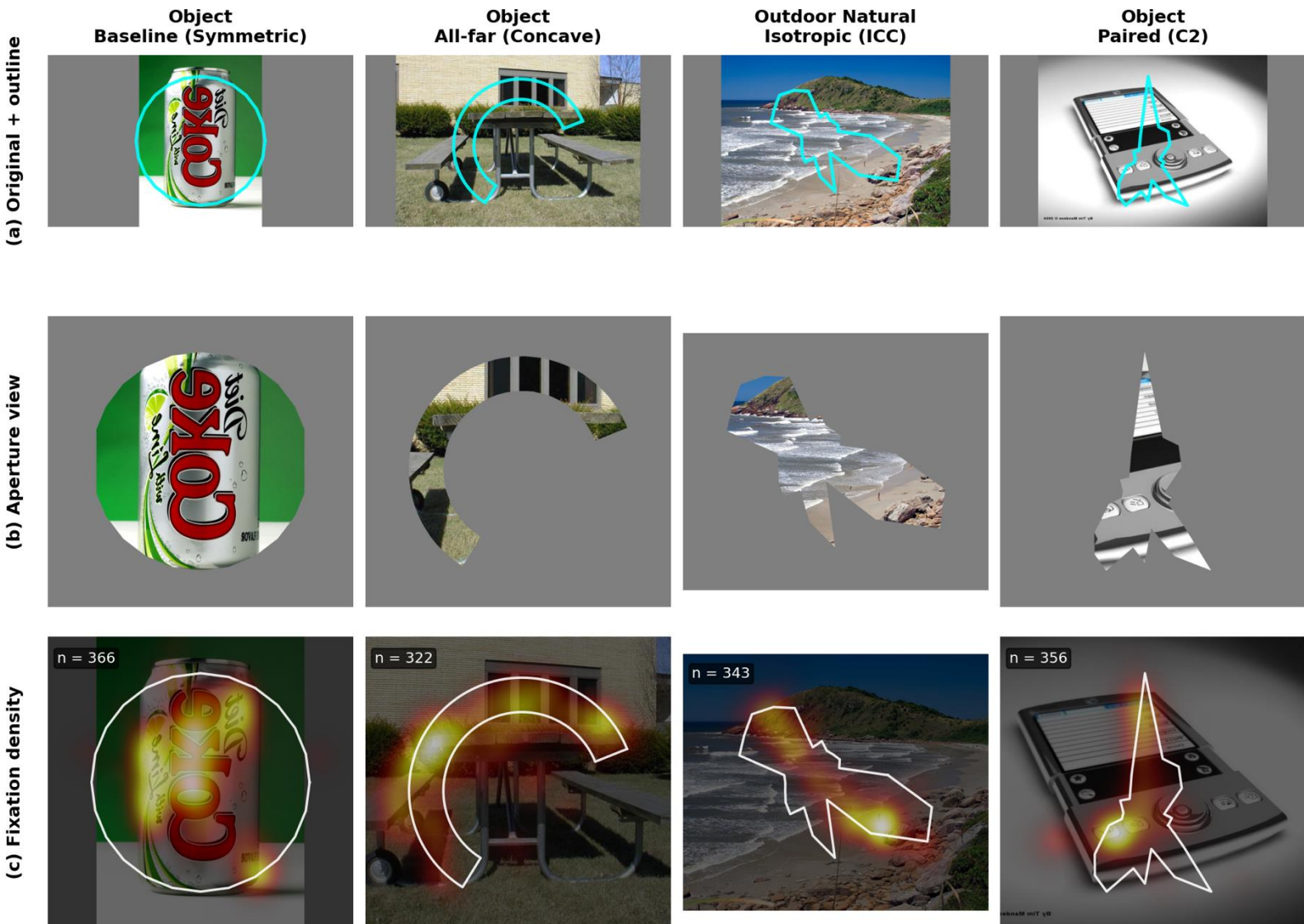
27 Polygonal Apertures

Reference baselines (1–3)
Rectangle + symmetric / asymmetric blob
Convexity-varied (4–6)
Convex · intermediate · concave
Irregular asymmetric (7–27)
7 configurations × 3 exemplars each



Polygons span symmetry, convexity, elongation, and boundary irregularity at comparable area.

Stimuli, Apertures & Fixations



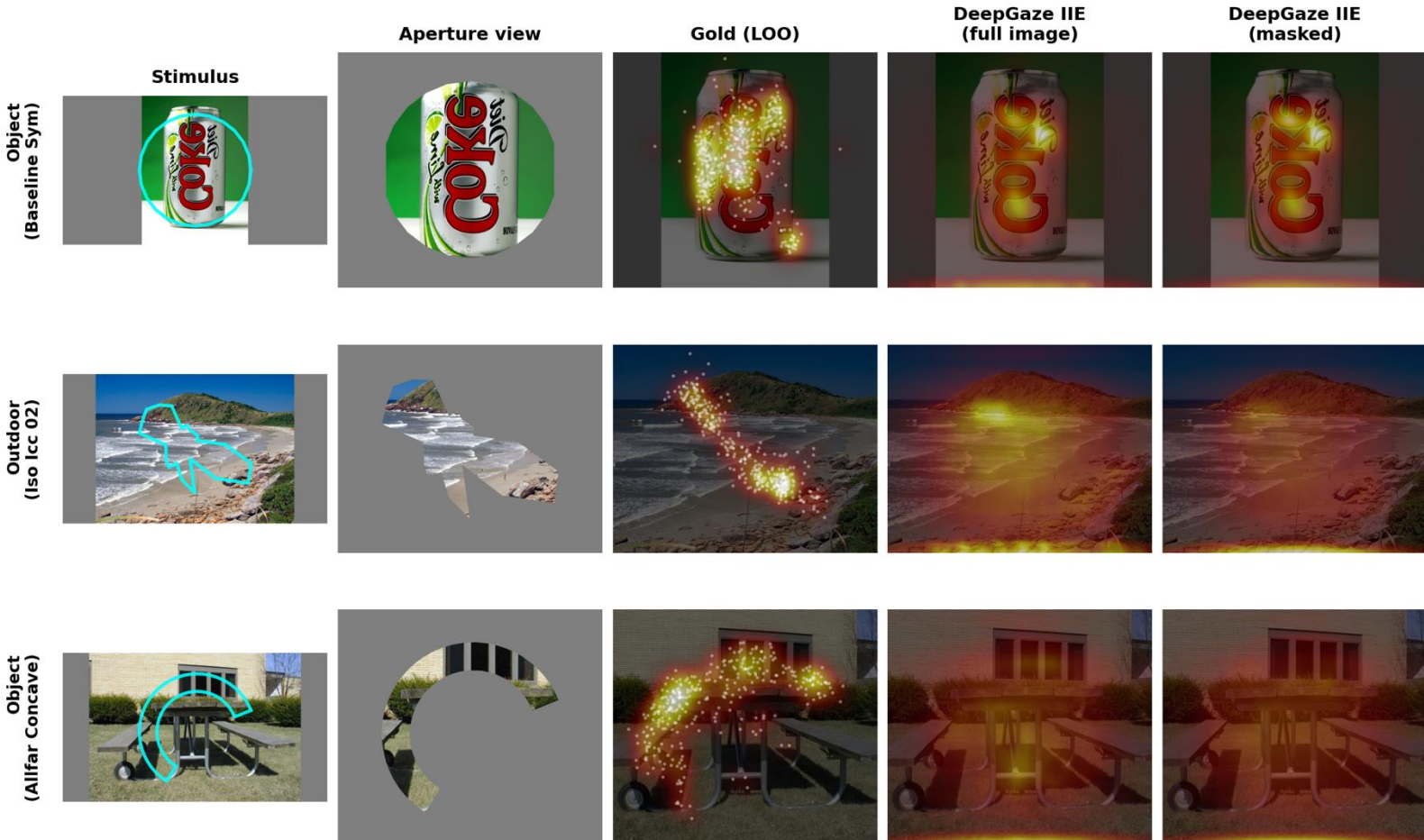
(a) original with polygon outline (cyan) · (b) aperture view shown to participants · (c) empirical fixation density across all observers.

Experimental Procedure



2 sessions × 9 mini-blocks × 39 trials (*702 trials/ participant*)
648 image-viewing + 54 empty-polygon trials per participant.
Attentive Viewing: Old/new memory probe after each mini-block, **mean accuracy 60%**

Saliency Predictions vs. Empirical Fixations



Stimulus · aperture · Gold (LOO) · DeepGaze IIE on full image · DeepGaze IIE on polygon-masked input.

Key Findings

- Center bias persists under apertures**
AUC = 0.91 — strong central tendency even when only the polygon region is visible.
- Inter-observer ceiling is preserved**
Gold (LOO) AUC = 0.97 — apertures don't disrupt cross-observer consistency.
- SOTA models transfer poorly**
DeepGaze IIE falls below Center Bias on NSS, CC, and Information Gain.
- Aperture geometry matters**
27 controlled shapes give experimental leverage beyond simple central tendency.

Released Artifacts

- Sample-level gaze streams**
TSV per session — apply any fixation algorithm.
- Fixation events**
CSV with timing, pixel & degree coords, in-polygon flag.
- Scanpath sequences**
Per-trial ordered fixations for temporal analyses.
- Empirical saliency maps**
NumPy + PNG · 648 polygon–image combinations.
- Polygon geometry**
27 JSON vertex files + geometry manifest.
- Quality metrics & splits**
Per-participant quality + image-level train/test (492/156).

Data Quality

97.2% valid trials (track ratio ≥ 0.80)	0.81 mean tracking ratio
0.6% mean data loss across participants	15 participants with zero data loss

Baseline Saliency Benchmark

Model	AUC↑	sAUC↑	NSS↑	CC↑	IG↑
Uniform	0.50	0.50	0.00	0.00	—
Center Bias	0.91	0.51	2.18	0.56	—
Gold (LOO)*	0.97	0.81	4.56	1.00	+1.60
DeepGaze IIE (full)	0.88	0.54	1.45	0.37	−0.58
DeepGaze IIE (masked)	0.89	0.57	1.68	0.41	−0.44

*Gold (LOO) is the inter-observer ceiling — excluded from model ranking.

Stimulus Coverage

600 images from CAT2000 · **6 categories × 100 images**

Fractal 100	Object 100	Outdoor Natural 100	Random 100	Satellite 100	Sketch 100
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Polygon–image pairs 648 unique · 30 observers each
Trial structure 702 / participant — 648 image + 54 empty
Train / Test split 492 / 156 pairs by image identity

Future Directions

- Aperture-aware saliency and scanpath models.
- Geometric drivers: symmetry, convexity, irregularity.
- Dynamic apertures — HMDs, occlusion, natural viewing.
- Task × aperture interaction (search vs. free viewing).
- Full CAT2000 coverage and cross-dataset transfer.

Get the Data

QR
placeholder
(GitHub Pages)

Scan for dataset, code, & project page

DOI 10.1145/3797246.3806795
HF laBran/PolyCAT
ICVL icvl.cs.bgu.ac.il/.../PolyCAT
License CC BY 4.0

CITE

Aklilu B. & Ben-Shahar O. (2026).
How do we look at images through polygonal apertures: The PolyCAT Dataset.
ETRA '26, Marrakesh, Morocco.

Quick Start

```
load_polycat.py

import pandas as pd
import numpy as np

# load fixations & a saliency map
fix = pd.read_csv("data/fixations/fixations_all.csv")
sal = np.load("data/saliency_maps/p12_i334.npy")

# inspect one trial
trial = fix[fix.trial_uid == "P05_A_p12_i334"]
print(trial[["x_px", "y_px", "duration_ms"]])
```

APPLY POLYCAT

research directions enabled by the dataset

Saliency Modeling

Train and benchmark aperture-aware saliency models
on 648 polygon–image pairs with standard MIT
metrics.

Scanpath Dynamics

Study how launch site, polygon boundary, and image
content interact to shape fixation sequences.

Geometric Vision

Probe how symmetry, convexity, and irregularity of
the visible region modulate gaze allocation.

Display & HMD Research

Use PolyCAT as a controlled proxy for HMDs,
occlusion, AR/VR — and extend toward dynamic
apertures.

Acknowledgements. We thank the members of the Interdisciplinary Computational Vision Laboratory (ICVL) for their support, and all participants who contributed their time to this study.

References. Itti & Koch (1998); Borji & Itti, CAT2000 (2015); Bylinskii et al. (2019); Linardos et al., DeepGaze IIE (2021); Akhtar et al., Croissant (2024); Mairon and Ben-Shahar (2025).

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